

HANGZHOU, China

WMO: 584570

Lat: 30.2259N

Lon: 120.1649E

Elev: 43

StdP: 100.81

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.0	(c) -0.8	(d) -12.8	(e) 1.3	(f) 3.8	(g) -10.0	(h) 1.6	(i) 3.9	(j) 7.6	(k) 4.9	(l) 6.6	(m) 5.1	(n) 2.0	(o) 320	(p) 0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.2	(c) 36.9	(d) 26.5	(e) 35.8	(f) 26.4	(g) 34.4	(h) 26.2	(i) 28.1	(j) 33.0	(k) 27.6	(l) 32.4	(m) 27.1	(n) 31.9	(o) 3.0	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.1	22.9	30.2	26.2	21.8	29.9	25.9	21.4	29.7	91.1	33.3	88.5	32.6	86.1	31.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.0	6.0	5.2	-4.2	38.6	1.6	1.2	-5.3	39.5	-6.3	40.2	-7.2	40.9	-8.4	41.7
			-5.5	29.1	1.6	0.8	-6.6	29.6	-7.6	30.0	-8.5	30.5	-9.7	31.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	5.2	7.1	11.5	17.3	22.3	25.4	29.9	29.3	24.9	19.7	13.5	7.5
	DBStd	9.03	3.32	3.91	4.13	3.77	2.86	2.84	2.78	2.55	3.03	2.88	3.70	3.39
	HDD10.0	384	154	98	30	1	0	0	0	0	0	0	11	91
	HDD18.3	1507	407	316	214	62	3	0	0	0	0	19	149	336
	CDD10.0	3254	5	16	77	221	382	462	616	598	447	299	116	13
	CDD18.3	1334	0	0	3	32	127	212	358	340	197	60	5	0
	CDH23.3	13608	1	6	37	234	867	1761	4687	4148	1597	248	24	0
	CDH26.7	6142	1	1	7	64	279	659	2531	2023	531	45	2	0
Wind	WSAvg	2.3	2.2	2.4	2.5	2.4	2.3	2.2	2.4	2.5	2.5	2.3	2.1	2.2
Precipitation	PrecAvg	1380	60	85	120	130	159	183	128	148	171	87	61	48
	PrecMax	2063	142	270	325	221	346	441	297	328	410	294	246	140
	PrecMin	955	0	12	32	42	83	48	28	13	38	1	2	0
	PrecStd	234	37	54	65	46	59	88	72	79	96	69	52	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	23.2	27.2	31.2	33.9	36.0	38.8	38.2	35.2	30.1	26.1	18.9
		MCWB	12.5	15.0	16.9	19.9	22.4	26.4	26.3	26.0	26.4	22.3	18.2	13.1
	2%	DB	14.8	18.5	23.5	28.5	31.8	34.0	37.1	36.6	33.1	27.9	23.1	16.2
		MCWB	9.7	12.8	15.3	19.0	20.8	25.8	26.5	26.5	25.6	20.6	16.7	11.7
	5%	DB	12.0	15.3	20.9	26.0	29.8	32.1	36.1	35.2	31.2	26.0	21.0	14.2
		MCWB	8.4	10.6	14.0	18.1	20.8	24.4	26.4	26.5	24.9	19.9	16.3	10.4
	10%	DB	10.0	12.8	18.1	23.9	27.9	30.2	35.0	34.0	29.8	24.2	19.1	12.8
		MCWB	7.3	9.1	13.1	17.0	20.1	24.0	26.3	26.5	24.1	18.7	15.4	9.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	16.2	19.1	22.8	25.4	27.7	29.0	28.9	28.2	24.7	20.8	15.3
		MCDB	18.1	21.2	23.7	27.2	30.9	32.3	34.1	33.8	32.2	27.9	23.1	16.9
	2%	WB	11.0	13.6	16.9	21.2	24.0	26.7	28.2	28.1	26.9	22.8	18.9	13.1
		MCDB	13.2	17.6	21.0	24.9	28.0	31.3	33.4	32.8	30.8	25.4	21.2	14.6
	5%	WB	9.1	11.4	15.2	19.9	23.0	26.0	27.7	27.5	26.0	21.5	17.5	11.6
		MCDB	11.0	14.4	19.4	23.8	26.6	30.3	32.8	32.1	29.5	24.0	19.6	13.4
	10%	WB	7.7	9.8	13.7	18.4	22.0	25.2	27.2	27.0	25.2	20.4	16.0	10.2
		MCDB	9.3	12.2	17.4	22.5	25.3	28.9	32.3	31.6	28.6	23.0	18.4	12.0
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.3	8.3	9.0	8.7	7.3	8.2	7.7	7.3	7.9	7.9	7.5
		MCDBR	11.2	13.1	13.4	13.2	12.2	10.2	9.8	9.7	9.1	10.1	11.1	10.9
		MCWBR	6.8	7.4	6.7	5.9	5.0	4.0	3.4	3.5	3.6	4.3	5.7	6.4
	5% WB	MCDBR	9.2	11.1	11.3	10.3	9.2	8.6	9.2	8.8	8.1	7.6	7.9	7.7
		MCWBR	6.3	6.9	6.6	5.8	4.9	4.0	3.8	3.6	3.7	4.2	4.9	5.4
Clear-Sky Solar Irradiance	taub	0.608	0.649	0.687	0.692	0.665	0.691	0.642	0.667	0.657	0.632	0.611	0.583	
	taud	1.647	1.586	1.550	1.591	1.684	1.671	1.799	1.724	1.714	1.694	1.689	1.698	
	Ebn at Noon	594	619	636	658	681	661	692	668	652	625	587	587	
	Edh at Noon	213	246	271	269	246	248	218	232	227	217	202	193	
All-Sky Solar Radiation	RadAvg	2.13	2.48	3.44	4.10	4.43	3.98	5.25	4.63	3.69	3.24	2.40	2.19	
	RadStd	0.35	0.57	0.29	0.50	0.53	0.60	0.63	0.60	0.38	0.50	0.35	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.56	N/A	N/A	N/A	-0.60	-0.86	N/A	-83	+152	+85
(47) Regional (14 neighbors)		+0.32	N/A	N/A	+0.52	N/A	N/A	N/A	N/A	+100	+65

Nomenclature: See separate page